```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
df = pd.read_csv('/content/titanic.csv', sep=r'\t')
```

```
/tmp/ipykernel_12378/283431005.py:1: ParserWarning: Falling back to the 'python' engine because the 'c' engine does not support
  df = pd.read_csv('/content/titanic.csv', sep=r'\t')
```

```
df
```

	"PassengerId Survived Pclass Lname Name Sex"
0	"1 0 3 Braund ...
1	"2 1 1 Cumings ...
2	"(Florence Briggs Thayer)"
3	"3 1 3 Heikkinen ...
4	"4 1 1 Futrelle ...
...	...
547	"S"
548	"S"
549	"S"
550	"C"
551	"🡅"

552 rows × 1 columns

```
df.head()
```

	"PassengerId Survived Pclass Lname Name Sex"
0	"1 0 3 Braund ...
1	"2 1 1 Cumings ...
2	"(Florence Briggs Thayer)"
3	"3 1 3 Heikkinen ...
4	"4 1 1 Futrelle ...

```
df.tail()
```

	"PassengerId Survived Pclass Lname Name Sex"
547	"S"
548	"S"
549	"S"
550	"C"
551	"🡅"

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 552 entries, 0 to 551
Data columns (total 1 columns):
 #   Column                                                                        Non-Null Count  Dtype
---  ------                                                                        --------------  -----
 0   "PassengerId        Survived       Pclass     Lname           Name              Sex"  552 non-null    object
dtypes: object(1)
memory usage: 4.4+ KB
```

```
df.columns
```

```
Index(['"PassengerId        Survived       Pclass      Lname           Name               Sex"'], dtype='object')
```

```
df.columns = df.columns.str.strip().str.lower()
```

```
df = pd.read_csv('/content/titanic.csv')
```

```
df = df[df.columns[0]].str.split(expand=True)
print(df.head())
```

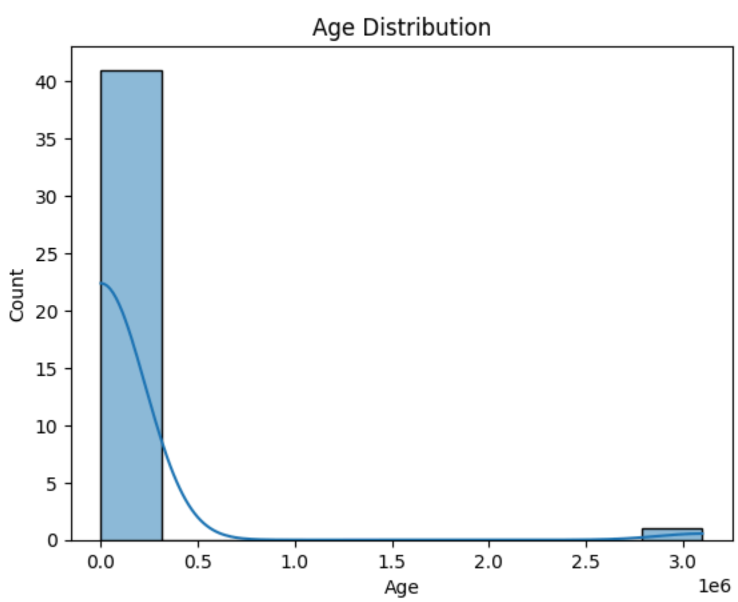
```
           0       1        2          3      4        5              6     7  \
0          1       0        3     Braund    Mr.     Owen         Harris  male
1          2       1        1    Cumings   Mrs.     John  Bradleyfemale  None
2  (Florence  Briggs  Thayer)       None   None     None           None  None
3          3       1        3  Heikkinen  Miss.    Laina         female  None
4          4       1        1   Futrelle   Mrs.  Jacques          Heath  None

      8
0  None
1  None
2  None
3  None
4  None
```

```
df.columns = ['PassengerId','Survived','Pclass','Name','Sex','Age','Fare','Cabin','Embarked']
```

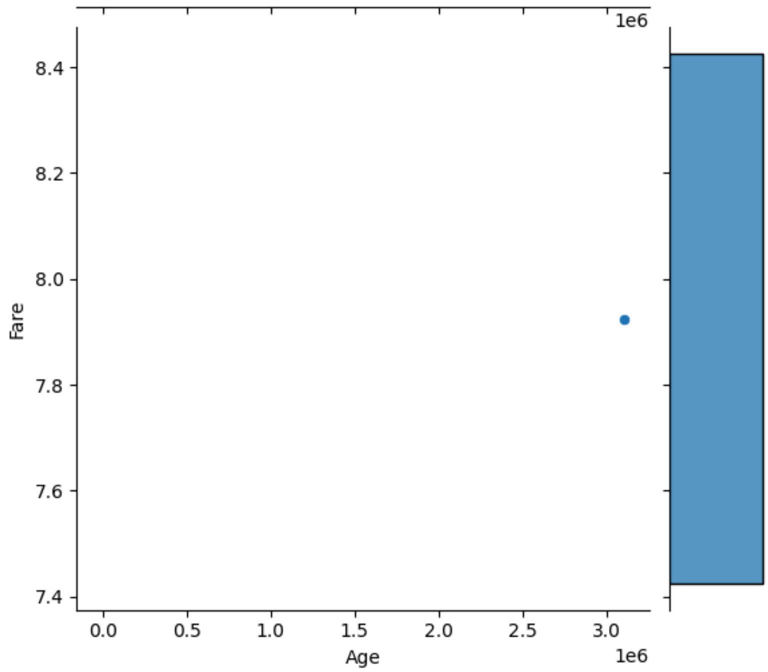
```
df['Age'] = pd.to_numeric(df['Age'], errors='coerce')
df['Fare'] = pd.to_numeric(df['Fare'], errors='coerce')
```

```
sns.histplot(df['Age'], bins=10, kde=True)
plt.title("Age Distribution")
plt.show()
```

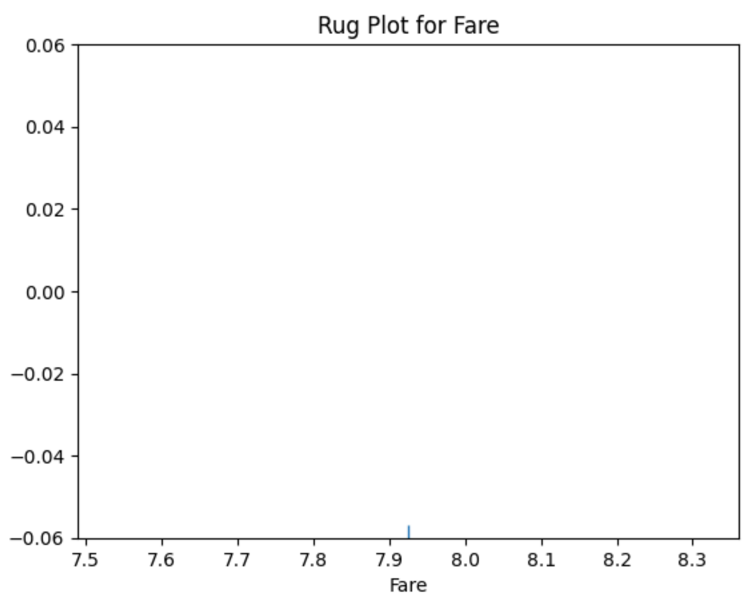



```
sns.jointplot(x='Age', y='Fare', data=df, kind='scatter')
```

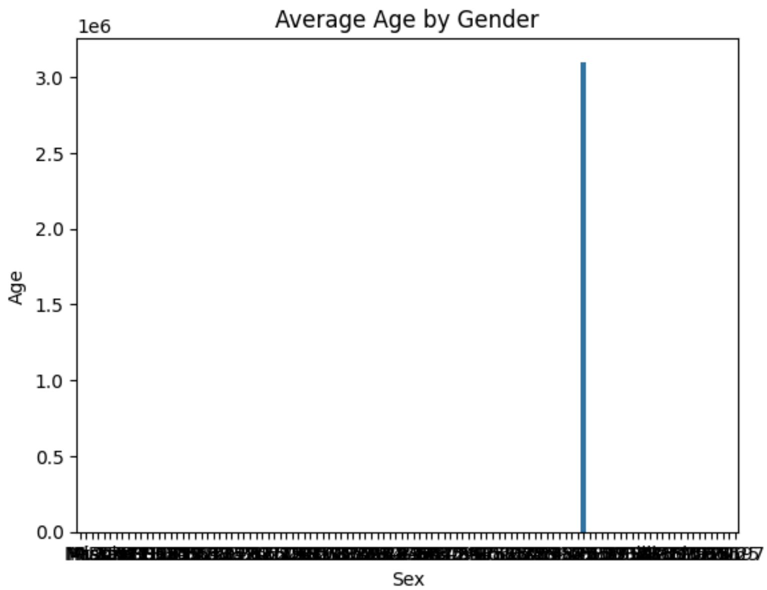
```
<seaborn.axisgrid.JointGrid at 0x7848b4c130b0>
```



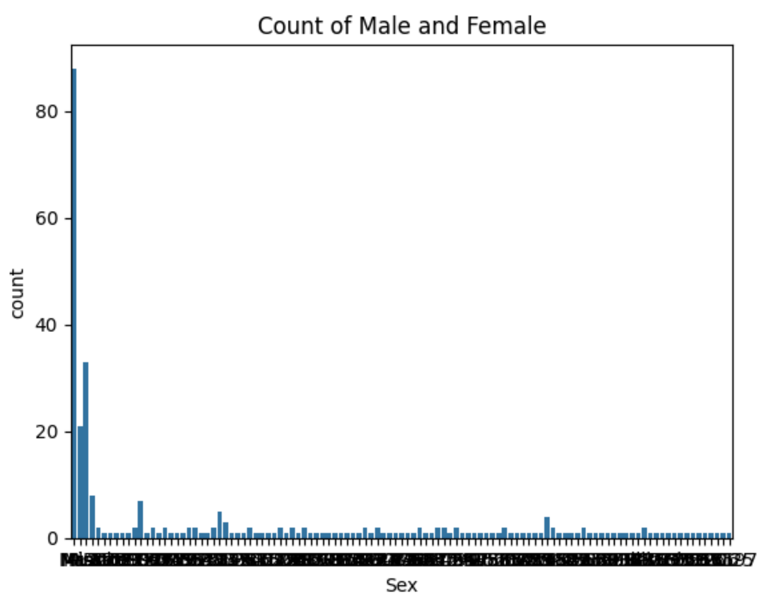

```
sns.rugplot(df['Fare'])
plt.title("Rug Plot for Fare")
plt.show()
```




```
sns.barplot(x='Sex', y='Age', data=df)
plt.title("Average Age by Gender")
plt.show()
```

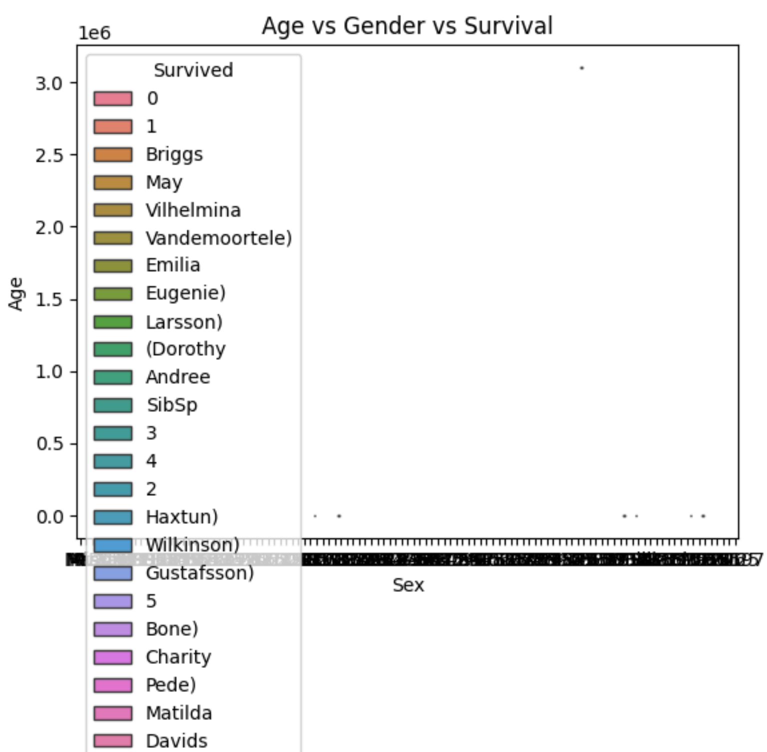



```
sns.countplot(x='Sex', data=df)
plt.title("Count of Male and Female")
plt.show()
```

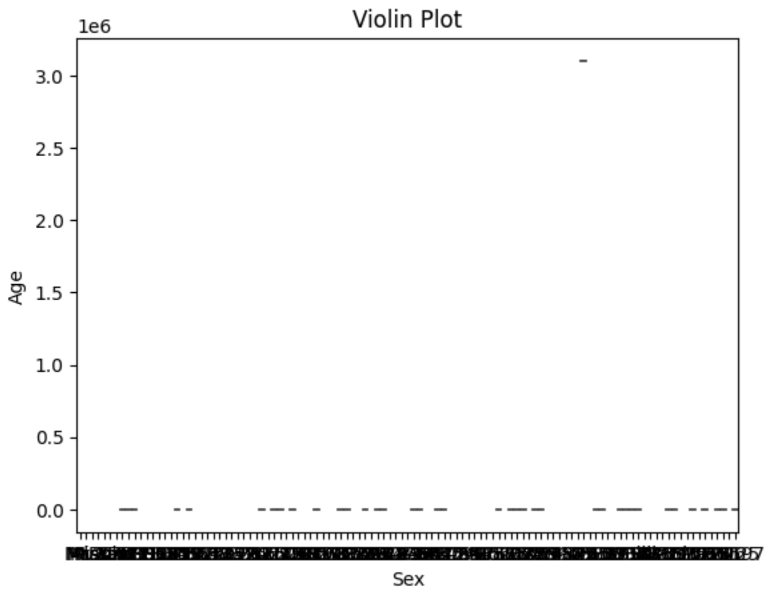



```
sns.boxplot(x='Sex', y='Age', data=df)
plt.title("Age Distribution by Gender")
plt.show()
```

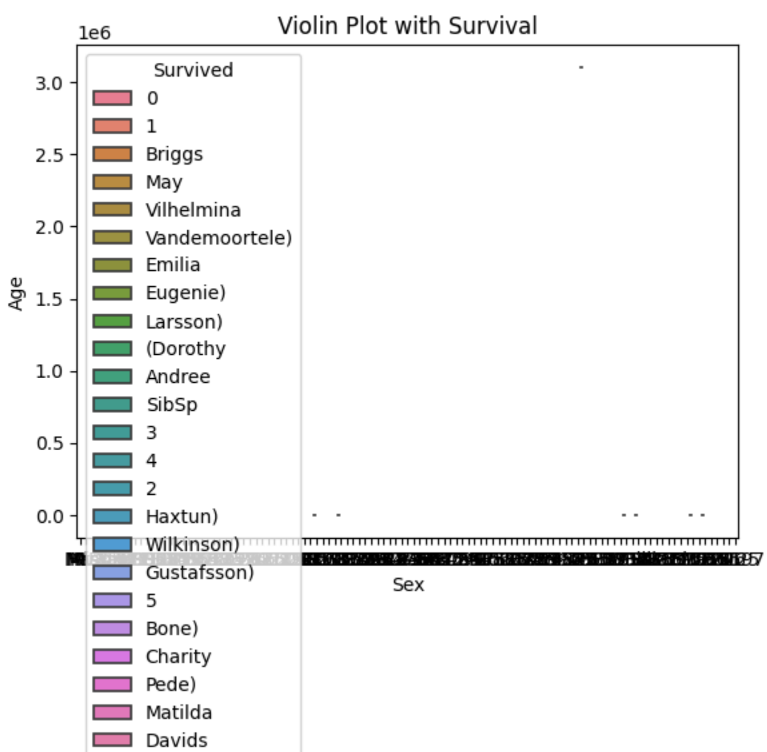

```
sns.boxplot(x='Sex', y='Age', data=df, hue='Survived')
plt.title("Age vs Gender vs Survival")
plt.show()
```



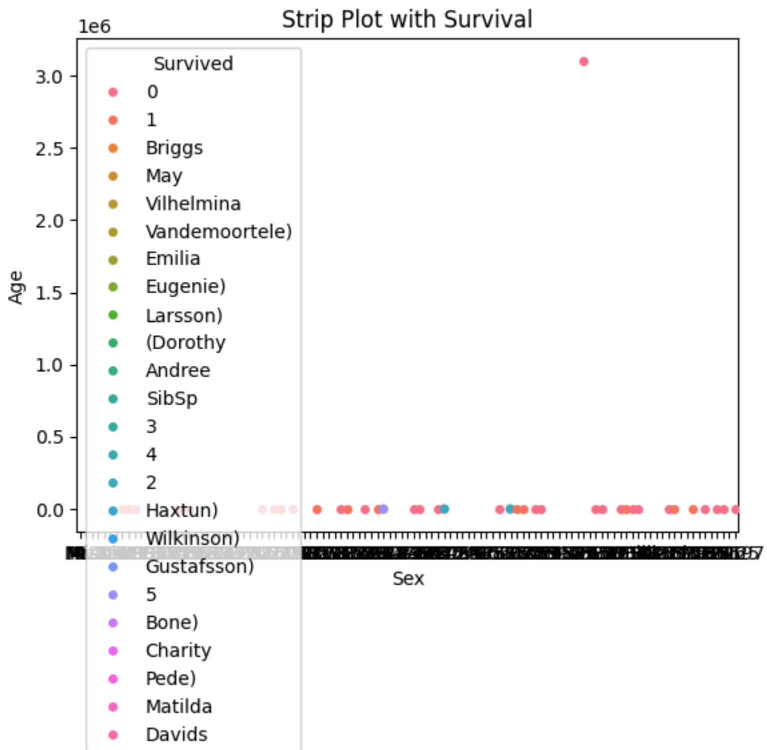

```
sns.violinplot(x='Sex', y='Age', data=df)
plt.title("Violin Plot")
plt.show()
```




```
sns.violinplot(x='Sex', y='Age', data=df, hue='Survived')
plt.title("Violin Plot with Survival")
plt.show()
```

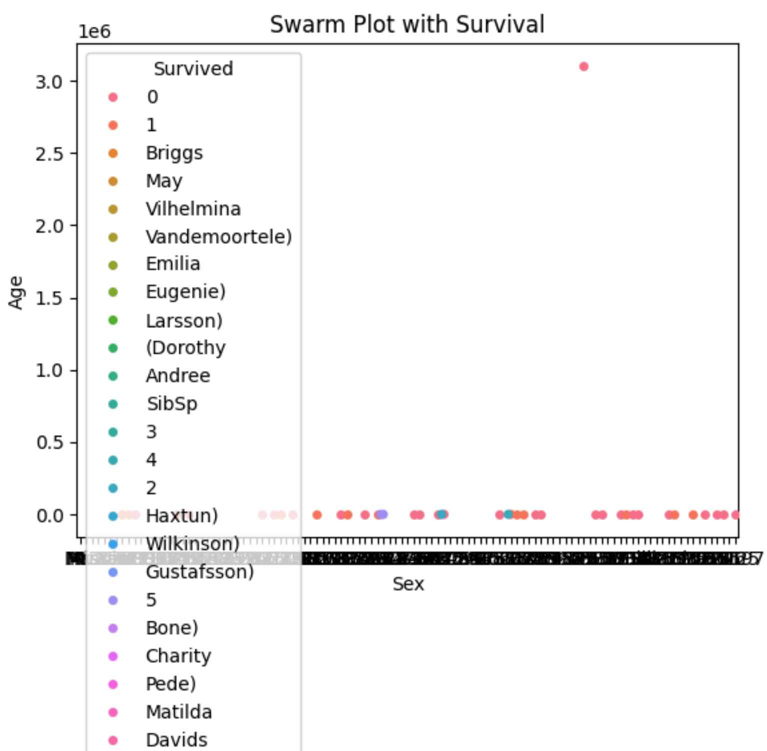



```
sns.stripplot(x='Sex', y='Age', data=df, jitter=True, hue='Survived')
plt.title("Strip Plot with Survival")
plt.show()
```



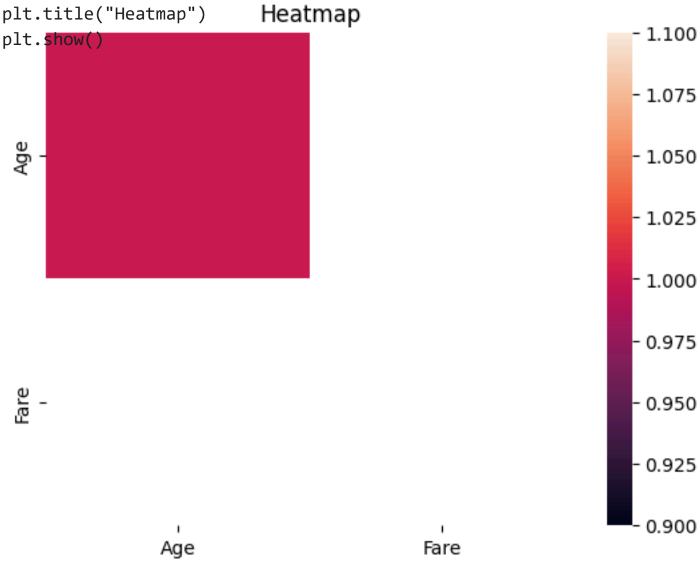

```
sns.swarmplot(x='Sex', y='Age', data=df, hue='Survived')
plt.title("Swarm Plot with Survival")
plt.show()
```

```
/usr/local/lib/python3.12/dist-packages/seaborn/categorical.py:3399: UserWarning: 50.0% of the points cannot be placed; you may
  warnings.warn(msg, UserWarning)
```




```
corr = df.corr(numeric_only=True)
```

```
sns.heatmap(corr)
plt.title("Heatmap")
plt.show()
```




```
sns.heatmap(corr, annot=True)
plt.title("Heatmap with Values")
plt.show()
```